

# Multi-source land-use emissions reveal rising airborne fraction

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**Abstract** The airborne fraction is the share of anthropogenic carbon dioxide emissions that remains in the atmosphere, and a key indicator of carbon-cycle response under continued emissions. Whether this share is rising remains debated because inference is sensitive to uncertainty in land-use and land-cover change (LULC) emissions. We address this by using all available LULC measurement series constructed from Global Carbon Budget 2025 data and estimating trends with a mixed-effects model with random intercepts and random slopes by LULC series. We find that airborne fraction has increased over time, and that this conclusion is robust to excluding the final year and to benchmarking against a measurement-error-aware weighted least squares estimator. These results strengthen evidence that an increasing share of emitted carbon dioxide is accumulating in the atmosphere rather than being taken up by land and ocean sinks, with direct implications for carbon-budget assessment and near-term mitigation requirements.

## 1 Background

2 Whether the airborne fraction (AF) is increasing or approximately constant remains contested<sup>1-7</sup>. In  
3 its classical form, AF is a yearly ratio of atmospheric growth to total anthropogenic emissions,  
4 computed as the sum of fossil fuel emissions and land-use and land-cover change emissions:

$$AF_t = \frac{G_t}{FF_t + LULC_t}, \quad (1)$$

5 where  $G_t$  is the annual atmospheric  $CO_2$  growth,  $FF_t$  is fossil fuel emissions excluding carbonation,  
6 and  $LULC_t$  is land-use and land-cover change emissions. AF is a key carbon-cycle diagnostic, with  
7 implications for carbon-cycle feedbacks and near-term mitigation planning<sup>1,2,8</sup>.

8 A persistent concern is that AF inference depends on the treatment of land-use and land-cover  
9 change (LULC) emissions, which are uncertain and model-dependent in annual carbon-budget

accounting. The Global Carbon Budget (GCB) 2025<sup>8</sup> provides one column of LULC emissions as the average of three bookkeeping models (BLUE<sup>9</sup>, OSCAR<sup>10</sup>, LUCE<sup>11</sup>), but a broader set of model-based LULC alternatives can be constructed from the same source data. If the information from this broader set of LULC measurements is not incorporated into AF trend inference, tests can be underpowered and inference on trend direction becomes less reliable.

Here we address that issue with a design that uses all repeated LULC measurements in a mixed-effects trend framework to obtain more reliable inference. The main specification estimates AF trends from a panel of yearly AF values constructed from each LULC measurement series, with random intercepts and random slopes capturing between-series heterogeneity and inducing within-series dependence over time. As a robustness exercise, we also estimate a two-stage measurement-error weighted least squares (WLS) model that propagates denominator uncertainty from repeated LULC measurements into annual AF variance and uses heteroskedasticity-and-autocorrelation-consistent (HAC) inference<sup>12–14</sup>.

Across specifications that incorporate the multi-source LULC information, we find robust evidence that AF has increased over 1959–2024, and that this conclusion is not driven by the final observation (2024), which shows a large AF jump. By contrast, ordinary least squares (OLS) specifications that do not incorporate denominator measurement information provide weaker evidence of a positive trend and are more sensitive to endpoint exclusion. Taken together, the results clarify why AF trend significance has been elusive and strengthen evidence that a growing share of emitted carbon dioxide is accumulating in the atmosphere.

## 30 **Data**

We use annual Global Carbon Budget 2025 data for 1959–2024, with atmospheric growth  $G_t$  from NOAA/ESRL global concentration trends<sup>15</sup>, fossil emissions excluding carbonation  $FF_t$  from the Global Carbon Project fossil dataset<sup>8</sup>, and a panel of 69 LULC measurements per year: BLUE<sup>9</sup>, OSCAR<sup>10</sup>, LUCE<sup>11</sup>, and peat-augmented process-based land-model combinations drawn from the GCB model ensemble<sup>16–36</sup>, combined with peat components<sup>37–39</sup> (see Methods). The data is shown in Figure 1.

Two LULC means constructed from the panel are shown. First, the GCB LULC mean, defined as the mean of the three bookkeeping models BLUE, OSCAR, and LUCE, corresponds to the LULC column used in the Global Carbon Budget. Second, the all-LULC mean, defined as the cross-series mean from all 69 LULC measurements. The literature has typically used the GCB LULC mean as the

denominator in AF trend estimation, but we show that incorporating the full set of LULC measurements materially changes inference on AF trends.

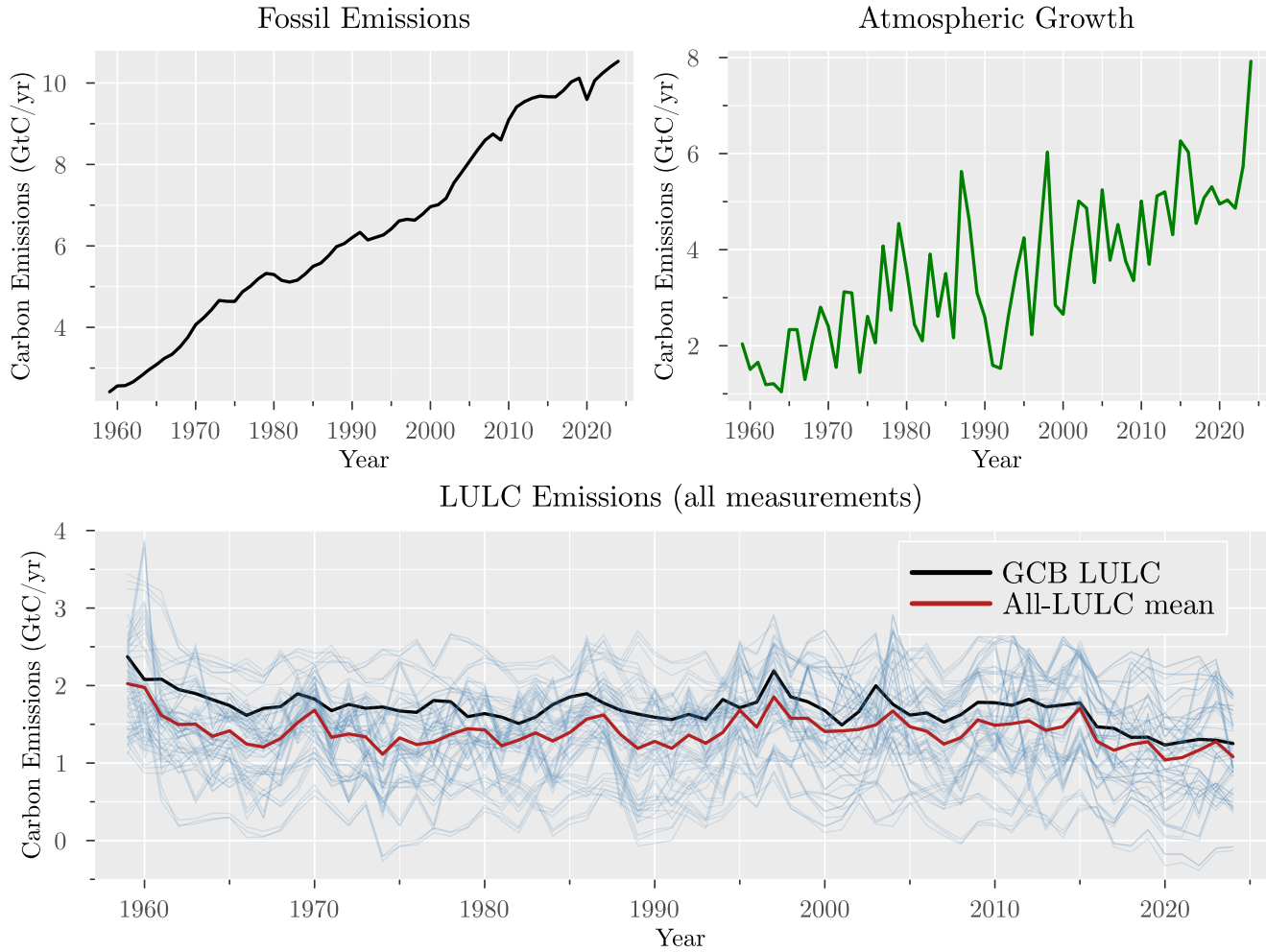


Figure 1: Global Carbon Budget 2025 data for fossil emissions, atmospheric growth, and LULC emissions (1959-2024). For the LULC panel, we show all extracted and derived LULC measurements together with the GCB LULC column and the all-LULC cross-series mean.

### Identification strategy

The empirical question is whether AF has a positive linear trend. The key design choice is how to use the full panel of repeated LULC measurements while accounting for within-series dependence and cross-series heterogeneity. OLS on a collapsed annual series discards panel information and hence provides less reliable inference, see results in Table 2. Our preferred specification is a mixed-effects model with random intercepts and random slopes by LULC series, allowing correlation across years within each series and capturing unobserved heterogeneity in LULC measurements<sup>40,41</sup> (see Methods).

For each LULC measurement series, we construct a series of AF estimates by year (Equation 1), and the model estimates a common time trend across all series while allowing for series-specific intercepts:

$$AF_{t,j} = \alpha_j + \beta_j t + \varepsilon_{t,j},$$

where

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \beta_j \sim N(\mu_\beta, \sigma_\beta^2), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

Here  $\mu_\alpha$  and  $\mu_\beta$  are the fixed effects (overall intercept and slope), while  $\sigma_\alpha^2$  and  $\sigma_\beta^2$  capture the variability in intercepts and slopes across LULC measurement series. The error term  $\varepsilon_{t,j}$  captures idiosyncratic noise. The key parameter of interest is  $\mu_\beta$ , which captures the average trend across all LULC measurement series.

## Results

### Main estimates

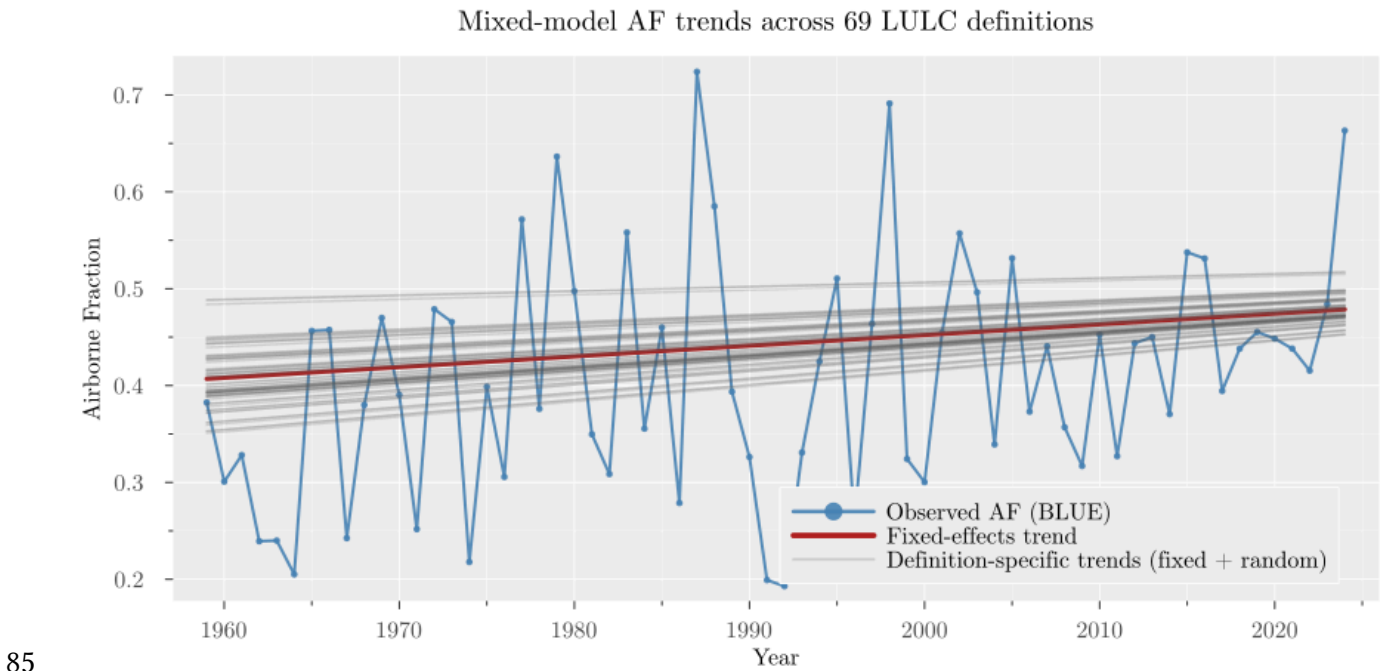
The main results of estimating the mixed-effects model considering the full sample (1959-2024) are shown in Table 1 and the associated trend lines are shown in Figure 2a.

Table 1: Mixed-effects model with random effects for each LULC measurement series.

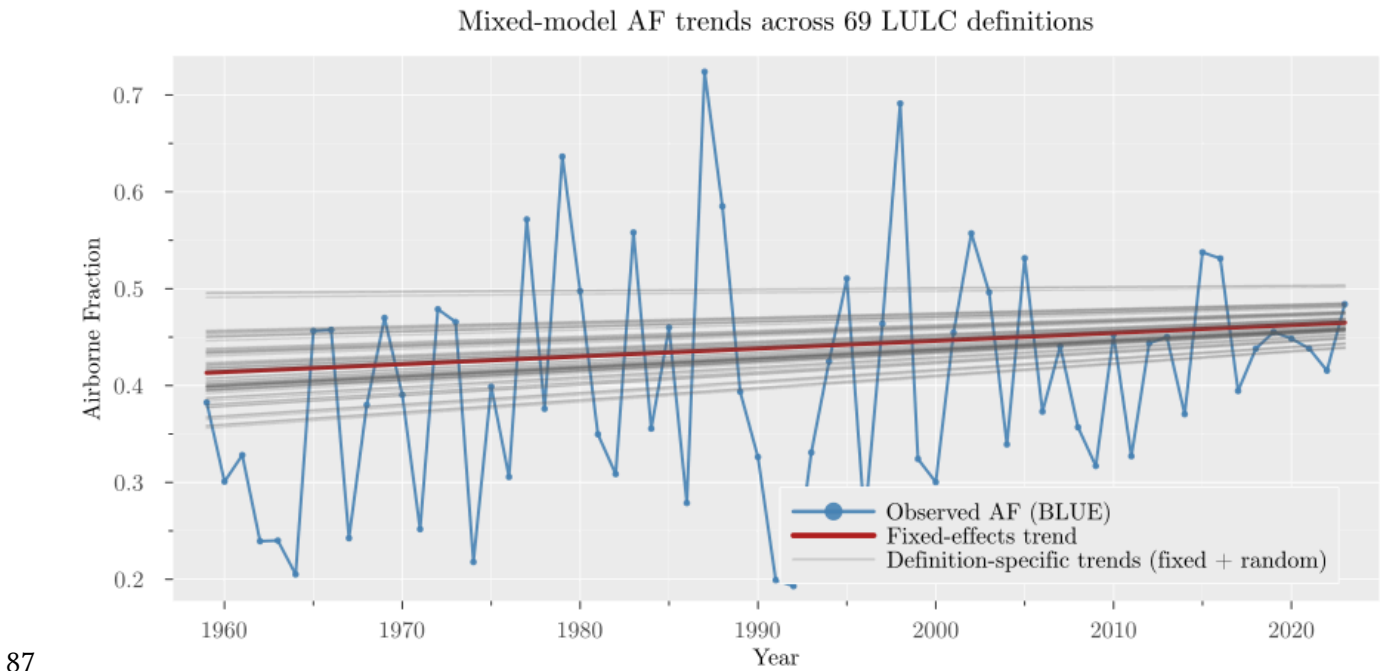
| Parameter               | Intercept (full) | Slope (full) | Intercept (up to 2023) | Slope (up to 2023) |
|-------------------------|------------------|--------------|------------------------|--------------------|
| Estimate                | 0.406959         | 0.001103     | 0.41326                | 0.000808           |
| Standard error          | 0.005573         | 0.000102     | 0.005586               | 0.000103           |
| p-value                 | 0.00000          | 0.00000      | 0.00000                | 0.00000            |
| R-squared (marginal)    | 0.008631         | 0.008631     | 0.004518               | 0.004518           |
| R-squared (conditional) | 0.700967         | 0.700967     | 0.708173               | 0.708173           |

The mixed-effects slope is positive and statistically significant (p-value < 1E-5), indicating strong evidence of an increasing AF trend across LULC measurement series. The estimated intercept is around 0.41 and the slope is around 0.0011 per year, implying an increase of about 0.07 in AF over 1959-2024. These estimates are comparable with the approximately 0.45 AF level reported in the literature<sup>2,3,6,7,42,43</sup>, while identifying a positive long-run trend when all LULC information is

82 incorporated. The conditional R-squared values indicate that most explained variation is captured by  
 83 between-series structure rather than by the fixed linear time trend (marginal versus conditional R-  
 84 squared).



85  
 86 (a) AF series with mixed-effects model trend (full sample, 1959-2024).



87  
 88 (b) AF series with mixed-effects model trend (sample ending in 2023).  
 89

Figure 2:

Figure 2a shows the AF series for each LULC measurement together with the overall mixed-effects trend line. The figure shows a clear positive trend across the panel of AF series, with the mixed-effects model capturing this common signal while allowing for series-specific heterogeneity.

### Endpoint and model specification robustness

AF shows a large jump in the final year (2024), so we re-estimate the mixed-effects model on the subsample ending in 2023 to check whether the result is driven by that observation. Results are shown in Table 1 and Figure 2b. The slope remains positive and statistically significant (p-value < 1E-5), indicating that the increasing AF trend is not driven by the final observation. The slope is smaller (0.000808 per year versus 0.001103 per year), consistent with a contribution from the 2024 jump, but the direction and inference are unchanged.

As additional robustness checks, we estimate a delta-method uncertainty proxy for each annual AF estimate using the cross-measurement dispersion of the full 69-series LULC panel, and use these proxies in WLS with HAC inference<sup>12–14,44</sup>. Two WLS specifications are shown in Table 2: one using the GCB LULC mean as the denominator and one using the all-LULC mean. In both WLS specifications in Table 2 weights are constructed from the all-LULC panel dispersion so that the weighting scheme is held fixed across denominator definitions. The WLS results are consistent with the mixed-effects results, and Supplementary Information confirms robustness to endpoint exclusion.

Table 2: Full-sample trend comparison across OLS and WLS specifications.

|                    | Trend, full sample | OLS (GCB) | OLS (all) | WLS (GCB) | WLS (all) |
|--------------------|--------------------|-----------|-----------|-----------|-----------|
| Estimate           |                    | 0.00158   | 0.00117   | 0.00265   | 0.00230   |
| Standard error     |                    | 0.00077   | 0.00079   | 0.00002   | 0.00002   |
| HAC standard error |                    | 0.00062   | 0.00064   | 0.00065   | 0.00066   |
| p-value            |                    | 0.04021   | 0.13891   | 0.00000   | 0.00000   |
| HAC p-value        |                    | 0.01112   | 0.06846   | 0.00004   | 0.00050   |
| R-squared          |                    | 0.06171   | 0.03309   | 0.14899   | 0.11107   |

By contrast, OLS specifications that do not incorporate denominator measurement information show weaker evidence of a positive trend and greater endpoint sensitivity. In the full sample, OLS using GCB LULC is significant under both conventional and HAC inference (p-values 0.04021 and 0.01112), whereas OLS using the all-LULC mean is not significant under either method (p-values 0.13891 and 0.06846). In the sample ending in 2023 (reported in the Supplementary Information), OLS using GCB LULC is only marginally significant under conventional inference (p-value = 0.09516), and OLS using

121 the all-LULC mean is not significant ( $p$ -value = 0.27109). This sensitivity to denominator construction  
122 and endpoint exclusion highlights the value of incorporating multi-source LULC information in AF  
123 trend estimation.

## 124 **Discussion and conclusion**

125 Using a framework that incorporates all available LULC information from Global Carbon Budget  
126 2025, we find robust evidence that AF increased from 1959 to 2024. Methodologically, incorporating  
127 multi-source uncertainty materially changes inference relative to plain OLS. Endpoint tests show that  
128 this conclusion is not driven by the 2024 jump.

129 A rising AF implies that a larger share of emitted  $CO_2$  remains in the atmosphere over policy-  
130 relevant horizons, tightening near-term mitigation requirements for a given temperature  
131 objective<sup>1,2,8</sup>. The increasing AF signal is consistent with broader evidence that the climate system is  
132 out of energy balance and has recently shown elevated warming and heating rates; these findings  
133 should be interpreted in that wider risk context<sup>45–48</sup>.

## 134 **Methods**

135 Our primary estimator is a linear mixed-effects trend model with random intercepts and random  
136 slopes for each LULC measurement series<sup>40,41,49,50</sup>. As a robustness check, we also estimate a two-stage  
137 measurement-error trend model using repeated yearly denominator measurements, followed by WLS  
138 trend estimation with HAC inference<sup>12–14,44,51</sup>.

## 139 **Construction of the LULC measurement panel**

140 The repeated LULC measurements are built from the Global Carbon Budget 2025 dataset. We first  
141 extract the three bookkeeping series (BLUE, OSCAR, LUCE)<sup>9–11</sup>. Then, for each process-based land-  
142 model LULC series in the GCB ensemble that does not already include peat emissions<sup>16–36</sup>, we add the  
143 corresponding peat component to make it comparable to the bookkeeping models, which include  
144 peat emissions by construction. Specifically, for each of the 22 process-based series we create three  
145 peat-augmented variants by adding `FAO_peat`, `LPX_Bern_peat`, and `ORCHIDEE_peat`<sup>37–39</sup>. This  
146 produces 66 derived series, and together with BLUE/OSCAR/LUCE gives a panel of 69 yearly LULC  
147 measurements.

## 148 **Mixed-effects model**

149 The primary estimator is a linear mixed-effects trend model fitted on the panel of yearly AF values  
150 constructed from each LULC measurement series. For each year  $t$  and series  $j$ , we define

$$AF_{t,j} = \frac{G_t}{FF_t + LULC_{t,j}},$$

151 and estimate

$$AF_{t,j} = (\mu_\alpha + u_{\alpha j}) + (\mu_\beta + u_{\beta j})t + \varepsilon_{t,j},$$

152 with

$$\begin{bmatrix} u_{\alpha j} \\ u_{\beta j} \end{bmatrix} \sim \left( N \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_u \right), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

153 This specification allows each LULC series to have its own level and trend while estimating a  
 154 common population-average trend  $\mu_\beta$ . The main estimand is  $\mu_\beta$ , interpreted as the average annual  
 155 change in AF across the full set of LULC definitions. Note that this notation is equivalent to the more  
 156 compact notation in the main text, where  $\alpha_j = \mu_\alpha + u_{\alpha j}$  and  $\beta_j = \mu_\beta + u_{\beta j}$ .

157 The model is estimated by restricted maximum likelihood to avoid bias in variance component  
 158 estimation<sup>40,41</sup>. Fixed-effect uncertainty is based on model-based standard errors and Wald tests. We  
 159 report the estimated fixed intercept and slope, standard errors, p-values, and marginal/conditional R-  
 160 squared<sup>49,50</sup>.

161 Identification and interpretation rely on standard mixed-model conditions: (i) conditional linearity in  
 162 time, (ii) random effects centered at zero and independent across LULC definitions, and (iii) mean-  
 163 zero residuals conditional on fixed and random effects.

164 Because shocks are common across all definitions within year, these shocks are absorbed into the  
 165 time-series component of the residual term; therefore, controlled-specification checks with additional  
 166 year-level covariates are treated as robustness analysis when reported.

167 As an endpoint robustness check, we re-estimate the same mixed-effects specification on the sample  
 168 ending in 2023 to verify that inference on  $\mu_\beta$  is not mechanically driven by the final-year  
 169 observation.

## 170 **Weighted least squares estimation approach**

171 As a robustness check to the mixed-effects model, we also estimate a two-stage measurement-error-  
 172 aware weighted least squares (WLS) model that uses repeated LULC measurements to construct a  
 173 denominator-uncertainty proxy for each year and uses that proxy for WLS weighting in a linear  
 174 trend regression with HAC inference<sup>12–14,44,51</sup>.

175 In the estimation, the yearly denominator is computed as



$$\hat{C}_t = FF_t + \overline{LULC}_t, \quad \overline{LULC}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} LULC_{tj},$$

176 where  $n_t = 3$  for the GCB LULC mean and  $n_t = 69$  for the all-LULC mean. In the implemented  
 177 robustness specification, the yearly weighting term is computed from the cross-measurement  
 178 dispersion across the full set of 69 LULC values and is then applied to both denominator definitions.  
 179 Assume for each time  $t = 1, \dots, T$  we observe:

- 180 • a single numerator  $b_t$  (e.g., atmospheric  $CO_2$  growth), and
- 181 • multiple denominator measurements  $c_{t1}, \dots, c_{tn_t}$  with  $n_t \geq 2$ , which are noisy observations of a  
 182 latent  $C_t$ .

183 Using the repeated denominator measurements, we construct a year-specific uncertainty proxy for  
 184 the ratio estimator  $a_t = b_t/C_t$  via the delta method, which captures how dispersion in denominator  
 185 definitions propagates into AF. The procedure is described in detail next.

186 We use a two-step approach to estimate the trend in the ratio  $a_t = b_t/C_t$  over time, accounting for  
 187 denominator measurement error.

188 **Step 1: Estimate  $C_t$  and  $a_t$**

- 189 1. Construct the denominator estimate at time  $t$  as the sample mean across available LULC  
 190 measurements:

$$\hat{C}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} c_{tj}.$$

191 We consider two variants of this denominator estimate: one using the GCB LULC column (the mean  
 192 of BLUE, OSCAR, LUCE) and one using the all-LULC mean (the mean across all 69 LULC  
 193 measurements).

194 The cross-measurement dispersion around  $\hat{C}_t$  is estimated from the multiple measurements.

195 With  $n_t \geq 2$ , the empirical dispersion estimator is

$$\widehat{\text{Disp}}(\hat{C}_t) = s_{c,t}^2, \quad s_{c,t}^2 = \frac{1}{n_t - 1} \sum_{j=1}^{n_t} (c_{tj} - \bar{c}_t)^2, \quad \bar{c}_t = \hat{C}_t.$$

- 196 2. Form the ratio estimate

$$\hat{a}_t = b_t / \hat{C}_t.$$

- 197 3. Use the multivariate delta method (shown below) to get an approximate variance of  $\hat{a}_t$

$$\widehat{\text{Var}}(\hat{a}_t) \propto \left( \frac{b_t}{\hat{C}_t^2} \right)^2 s_{c,t}^2.$$

198 **Step 2: WLS regression of  $\hat{a}_t$  on time**

199 For each time  $t$ , we obtain a point estimate  $\hat{a}_t$  and an uncertainty proxy that reflects cross-  
 200 measurement dispersion. We use these quantities to fit a linear trend via weighted least squares  
 201 (WLS):

$$\hat{a}_t = \alpha + \beta t + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2), \quad \sigma_t^2 = \widehat{\text{Var}}(\hat{a}_t).$$

202 In our application, the time index  $t$  is the year, and  $\hat{a}_t$  is the estimated AF for that year. The  
 203 weighting term  $\sigma_t^2$  is proportional to estimated AF uncertainty from denominator dispersion and  
 204 varies across years with the spread of the LULC measurement panel. We use this term to weight the  
 205 regression, giving more weight to years with lower denominator-related uncertainty.

206 Assuming negligible correlation in  $\varepsilon_t$  across time, this is WLS with weights  $w_t = \frac{1}{\sigma_t^2}$ . If we suspect  
 207 serial correlation, we can use HAC standard errors for inference on  $\beta$  without changing the point  
 208 estimate. In practice, we use a HAC covariance estimator with a Bartlett kernel and Andrews  
 209 automatic bandwidth selection<sup>14</sup>. In the results, we report both conventional and HAC standard  
 210 errors.

211 Testing for a time trend is then a test of  $\beta = 0$  versus  $\beta \neq 0$  in this linear model, and WLS uses the  
 212 repeated denominator information through this delta-method weighting scheme.

213 **Delta method for ratio variance estimation**

214 **Step-by-step derivation (first-order delta method):**

215 Define the random vector and the function of interest as

$$X_t = \begin{bmatrix} b_t \\ \hat{C}_t \end{bmatrix}, \quad g(X_t) = \frac{b_t}{\hat{C}_t}.$$

216 and let

$$\sigma_{b,t}^2 = \text{Var}(b_t), \quad \sigma_{C,t}^2 = \text{Var}(\hat{C}_t), \quad \sigma_{bC,t} = \text{Cov}(b_t, \hat{C}_t).$$

217 1. Linearize  $g$  around the mean vector  $(\mu_{b,t}, \mu_{C,t}) = \mathbb{E}[(b_t, \hat{C}_t)]$ :

$$g(X_t) \approx g(\mu_{b,t}, \mu_{C,t}) + \nabla g(\mu_{b,t}, \mu_{C,t})^\top (X_t - \mathbb{E}[X_t]).$$

218 2. Compute the gradient:

$$\nabla g(b, C) = \begin{bmatrix} \partial g / \partial b \\ \partial g / \partial C \end{bmatrix} = \begin{bmatrix} 1/C \\ -b/C^2 \end{bmatrix}.$$

219 3. Write the covariance matrix of  $(b_t, \hat{C}_t)$ :

$$\Sigma_t = \begin{bmatrix} \sigma_{b,t}^2 & \sigma_{bC,t} \\ \sigma_{bC,t} & \sigma_{C,t}^2 \end{bmatrix}.$$

220 4. Apply the delta-method variance formula

$$\text{Var}(g(X_t)) \approx \nabla g(\mu_{b,t}, \mu_{C,t})^\top \Sigma_t \nabla g(\mu_{b,t}, \mu_{C,t}),$$

221 and with plug-in evaluation at  $(b_t, C_t)$ , this becomes

$$\text{Var}(\hat{a}_t) \approx \left(\frac{1}{C_t}\right)^2 \sigma_{b,t}^2 + \left(\frac{b_t}{C_t^2}\right)^2 \sigma_{C,t}^2 - 2\frac{b_t}{C_t^3} \sigma_{bC,t}.$$

222 In practice, we use plug-in estimates (replace unknown moments by their empirical counterparts).

223 This is the standard delta-method approximation for the variance of a ratio estimator.

224 5. In the case analysed in this paper, we have a single numerator measurement from a robust source

225 per time, so we treat  $\sigma_{b,t}^2$  and  $\sigma_{bC,t}$  as negligible relative to the variance from the denominator

226 measurement error, which is captured by  $\sigma_{C,t}^2$ .

227 The expression simplifies to

$$\text{Var}(\hat{a}_t) \approx \left(\frac{b_t}{C_t^2}\right)^2 \sigma_{C,t}^2.$$

228 Accordingly, a practical plug-in weighting proxy is

$$\tilde{\sigma}_{a,t}^2 = \left(\frac{b_t}{\hat{C}_t^2}\right)^2 s_{c,t}^2.$$

## 229 **Data availability**

230 The study uses publicly available Global Carbon Budget 2025 data and derived yearly series

231 generated from those source files. The processed analysis tables are provided in the manuscript

232 repository under the results directory, and the extracted and derived LULC panel is available as a

233 CSV file in the data directory.

## 234 **Code availability**

235 All code used to process data, estimate models, and generate figures is available in the repository

236 ([github.com/everval/Airborne-Fraction-WLS-Trend](https://github.com/everval/Airborne-Fraction-WLS-Trend)) under the scripts directory, including Quarto

237 analysis files and Julia helper functions.

## 238 **Competing interests**

239 The author declares no competing interests.

## 240 **Additional information**

241 Supplementary information is available in the accompanying supplementary manuscript file.

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